Vishva Niral Patel

Electronics & Communication Engineering | Hardware, Embedded Systems, Wireless Research & Intelligent Systems

Gujarat, India | +91-9773285975 | vishvapatel.ece@gmail.com | vishvaprojects.in | github.com/VishvaPatelECE | linkedin.com/in/vishva-patel-ece

Education

Dharmsinh Desai University (DDU) — Nadiad, Gujarat, India

Bachelor of Technology (B.Tech.) in Electronics and Communication Engineering — *2023 – Expected May 2027*

- CGPA: **8.99** / 10.0
- Higher Secondary (Class XII, Science), GHSEB: **78.3%** (2023)
  Secondary (Class X), GHSEB: **88.5%** (2021)

Research Experience

Sardar Vallabhbhai National Institute of Technology (SVNIT) — Surat, India

AI Research Intern, Department of Artificial Intelligence — *May 2026 to July 2026*

GraphMARL: Graph-Based Resource Allocation for Vehicular Wireless Networks

- Designed and integrated a Python C-V2X simulator (selected 3GPP TR 38.901 path loss, correlated shadowing, Rayleigh fading, co-channel interference) with SUMO/TraCI traffic to train and test graph-based multi-agent reinforcement learning.
- In the evaluated setup, paired-seed tests found no statistically significant throughput difference between the learned policy and state-matched random allocation ($\mathbf{p = 0.824}$). A one-hop sidelink coordination rule raised throughput by about **10.8% to 11.7%** for both policies; broader testing across regimes is needed to establish when learning adds value.

Core Engineering Projects

Code repositories will be made available on request.

RANGraph: Semantic Compilation for Radio Access Network Telemetry | *Python, Rust* — 2026

- Architected a streaming telemetry compiler parsing heterogeneous cellular observations into a typed Canonical IR, enforcing strict dimensional unit safety (dBm vs. linear power) and resolving PCI collisions via global ECGI mapping.
- Engineered an incremental DAG execution engine with dependency-aware selective recomputation, restricting recalculation strictly to mutated downstream subgraphs while eliminating redundant full-graph evaluation.
- Verified complete output equivalence against an independent clean-room recomputation oracle across **2,772** factorial benchmark trials (22 configs $\times$ 3 seeds $\times$ 7 reps $\times$ 6 systems), achieving 100% oracle parity under numerical tolerance $\Delta < 5 \times 10^{-4}$ with zero stale or divergent outputs.
- Passed **250/250** adversarial contract tests across 5 structural failure families, outperforming Relational ETL (26.0%), RDF/SHACL (52.0%), and 3GPP standards (48.0%); achieved **0.080 ms** median latency at 100,000 nodes (**2,776**$\times$ speedup over cold scratch); recorded **22.7**$\times$ (Vienna) and **18.2**$\times$ (UCCMISL) trace speedups; noted dense batches ($k \geq 500$) favor Differential Dataflow.

CMEL: Contextual Marginal Externality Logic (derived from GraphMARL research) | *V2X scheduling, SUMO* — 2026

- Designed a single-pass, urgency-ordered scheduling algorithm for interference-constrained vehicular wireless networks, prioritizing packets by latency deadlines and channel decodability indicators.
- Formulated closed-form Huang-Berry-Honig (HBH) marginal interference pricing to allocate PRBs, evaluating delivery rate gains against aggregate network externality in a single pass without iterative game rounds.
- Achieved a **+4.19 percentage-point** increase in on-time packet delivery over TimelyGreedy across **180** sparse-to-moderate evaluation scenarios ($N \leq 48$, **180 wins, 0 losses**; 95% CI: **+3.88 to +4.49** percentage points; $p < 10^{-5}$); matched 25-round iterated HBH game pricing quality in a single non-iterative pass.
- Identified density-load operating boundaries: delivery was **1.63 percentage points** lower under dense traffic ($N \geq 72$) because continuous rate models diverged from thresholded packet decoding cliffs under severe co-channel interference.

DARVS: Biometric Verification with Challenge-Response Liveness | *Python, OpenCV, Flask, MTCNN, FaceNet* — 2025

- Built an end-to-end identity authentication system integrating MTCNN facial detection and 512-dimensional FaceNet feature embeddings with dynamic 2D challenge-response movement verification to separate physical liveness from identity recognition.
- Designed an interactive challenge-response state machine issuing randomized prompt sequences (head shifts $\geq 7\%$, head tilts $\geq 5\%$) requiring 2-frame persistence, coupled with a 3-frame rolling average on facial similarity scores (≥ 0.78 threshold).
- Trained and evaluated a custom 84K-parameter CNN eye-state classifier, achieving **87.9%** binary accuracy and **95.2%** closed-eye recall on a localized 620-image development dataset (disclosed single-subject validation set, not an external academic benchmark).
- Engineered a modular Flask REST backend with SQLite credential storage, webcam video streaming, and automated event audit logging; successfully mitigated static 2D photo and display spoofs across 41 logged prototype verification attempts.

USMHACS: Hardware Access Control System | *Digital logic, Proteus* — 2025

- Designed, simulated, and physically constructed a microcontroller-free access-control system utilizing dual cascaded 74LS138 3-to-8 decoders to decode a 4-bit parallel hardwired DIP passcode.
- Engineered an asynchronous fail-secure tamper memory unit using a CD4043B quad NOR SR latch and CD4068 8-input AND logic, establishing irreversible hardware lockout upon intrusion detection until an authorized dual-condition reset is asserted.
- Implemented an NE555 timer in monostable configuration ($R = 10\ \text{k}\Omega, C = 470\ \mu\text{F}$) generating an exact **5.17-second** solenoid unlock pulse, interfaced with a 74HC14 hex Schmitt trigger and RC networks for contact debouncing.
- Validated 8 comprehensive operational and fault scenarios through Proteus 8.9 SP2 simulation and physical breadboard hardware; presented the working implementation at the **Eureka Technical Exhibition 2025**.

Technical Skills

Programming & Systems: Python, C, C++, Rust, Verilog HDL, Scilab, Bash, SQL, Linux/Unix CLI

Wireless & Telecommunications: C-V2X (Mode 4), 3GPP TR 38.901 & TR 37.885 Channel Modeling, PRB Scheduling, SINR & Interference Modeling, Huang-Berry-Honig Pricing, SUMO/TraCI Mobility

Digital Hardware & VLSI: Digital Logic Design, Finite State Machines (FSMs), RTL Design, 74-Series TTL, CD4000 CMOS Logic, Hardware Security, CMOS VLSI Fundamentals, Proteus 8.9, NI Multisim, Quartus II, ModelSim, Cadence (Virtuoso & IMC)

Embedded Systems: ESP32, ARM Cortex-M3 (Keil μVision), 8051 Microcontroller, 8086 / ARM7TDMI Microprocessors, Sensor Interfacing (IR, Ultrasonic, RFID), Protocols: I2C, SPI, UART

Machine Learning & Computer Vision: PyTorch, PyTorch Geometric (PyG), TensorFlow/Keras, OpenCV, MTCNN, FaceNet, Custom CNNs, Multi-Agent Reinforcement Learning (MAPPO, MASAC), Graph Neural Networks (GNNs)

Research & Verification Methods: Paired-Seed Benchmarking, Statistical Hypothesis Testing (t-tests, Wilcoxon), Ablation Studies, Clean-Room Oracle Verification, Adversarial Contract Testing, Git, LaTeX

Publications (Manuscripts in Preparation)

- V. Patel, "When Interference Pricing Meets Thresholded Timely Delivery: A Density-Load Operating Boundary in V2X."
- V. Patel, "RANGraph: Semantic Compilation for Radio Access Network Telemetry."

Relevant Coursework

Communication & Signals: Signals and Systems, Digital Signal Processing, Analog & Digital Communication, Wireless Communication, Data & Computer Communication, Information Theory, Coding Theory and Compression

Electronics & Hardware: Digital Electronics, Linear Electronics I & II, CMOS VLSI Design, Electromagnetic Fields, Microwave & Antennas

Embedded & Computing: Microcontroller Applications (8051), Advanced Microprocessors (8086, ARM7TDMI), Embedded Systems (ARM Cortex-M3), Control Systems, Computer Vision and Machine Learning

Mathematics & Analytics: Linear Algebra, Probability Theory and Random Processes, Discrete Mathematics, Statistical Inference

Honors, Awards & Leadership

GATE EC 2026: All India Rank (AIR): **6,931** | GATE Score: **451** 2026

Eureka Technical Exhibition (2025): Presented USMHACS (Hardware Access-Control System) 2025

Aspire Leaders Program (2025): Leadership and ethics curriculum founded by Harvard Business School faculty 2025

Research Interests & Languages

Research Interests: Wireless Communication Systems, Domain-Specific Compilers, Sidelink Resource Allocation, Discrete Hardware Architectures, Embedded Systems, Computer Vision

Languages: English (Professional Working Proficiency), Hindi (Fluent), Gujarati (Native)